

Project Euler Solutions

Claude Opus 4.8

Contents

Problem 1: Multiples of 3 or 5	4
Problem 2: Even Fibonacci Numbers	5
Problem 3: Largest Prime Factor	6
Problem 4: Largest Palindrome Product	7
Problem 5: Smallest Multiple	8
Problem 6: Sum Square Difference	9
Problem 7: 10001st Prime	10
Problem 8: Largest Product in a Series	11
Problem 9: Special Pythagorean Triplet	12
Problem 10: Summation of Primes	13
Problem 11: Largest Product in a Grid	14
Problem 12: Highly Divisible Triangular Number	15
Problem 13: Large Sum	16
Problem 14: Longest Collatz Sequence	17
Problem 15: Lattice Paths	18
Problem 16: Power Digit Sum	19
Problem 17: Number Letter Counts	20
Problem 18: Maximum Path Sum I	21
Problem 19: Counting Sundays	22
Problem 20: Factorial Digit Sum	23
Problem 21: Amicable Numbers	24
Problem 22: Names Scores	25
Problem 23: Non-Abundant Sums	26
Problem 24: Lexicographic Permutations	27
Problem 25: 1000-digit Fibonacci Number	28
Problem 26: Reciprocal Cycles	29
Problem 27: Quadratic Primes	30
Problem 28: Number Spiral Diagonals	31
Problem 29: Distinct Powers	32
Problem 30: Digit Fifth Powers	33
Problem 31: Coin Sums	34
Problem 32: Pandigital Products	35
Problem 33: Digit Cancelling Fractions	36
Problem 34: Digit Factorials	37
Problem 35: Circular Primes	38
Problem 36: Double-base Palindromes	39
Problem 37: Truncatable Primes	40
Problem 38: Pandigital Multiples	41
Problem 39: Integer Right Triangles	42
Problem 41: Pandigital Prime	43
Problem 42: Coded Triangle Numbers	44

Problem 43: Sub-string Divisibility	45
Problem 44: Pentagon Numbers	46
Problem 46: Goldbach's Other Conjecture	47
Problem 47: Distinct Primes Factors	48
Problem 48: Self Powers	49
Problem 49: Prime Permutations	50
Problem 50: Consecutive Prime Sum	51
Problem 51: Prime Digit Replacements	52
Problem 52: Permuted Multiples	53
Problem 53: Combinatoric Selections	54
Problem 55: Lychrel Numbers	55
Problem 56: Powerful Digit Sum	56
Problem 57: Square Root Convergents	57
Problem 58: Spiral Primes	58
Problem 62: Cubic Permutations	59
Problem 63: Powerful Digit Counts	60
Problem 65: Convergents of e	61
Problem 67: Maximum Path Sum II	62
Problem 69: Totient Maximum	63
Problem 70: Totient Permutation	64
Problem 71: Ordered Fractions	65
Problem 72: Counting Fractions	66
Problem 73: Counting Fractions in a Range	67
Problem 74: Digit Factorial Chains	68
Problem 76: Counting Summations	69
Problem 77: Prime Summations	70
Problem 80: Square Root Digital Expansion	71
Problem 85: Counting Rectangles	72
Problem 87: Prime Power Triples	73
Problem 92: Square Digit Chains	74
Problem 95: Amicable Chains	75
Problem 97: Large Non-Mersenne Prime	76
Problem 99: Largest Exponential	77
Problem 104: Pandigital Fibonacci Ends	78
Problem 108: Diophantine Reciprocals I	79
Problem 112: Bouncy Numbers	80
Problem 114: Counting Block Combinations I	81
Problem 115: Counting Block Combinations II	82
Problem 116: Red, Green or Blue Tiles	83
Problem 118: Pandigital Prime Sets	84
Problem 123: Prime Square Remainders	85
Problem 124: Ordered Radicals	86
Problem 125: Palindromic Sums	87
Problem 145: Reversible Numbers	88
Problem 173: Square Laminae	89
Problem 179: Consecutive Positive Divisors	90

Problem 187: Semiprimes	91
Problem 188: Hyperexponentiation	92
Problem 203: Squarefree Binomial Coefficients	93
Problem 301: Nim	94
Problem 315: Digital Root Clocks	95
Problem 323: Bitwise-OR Operations on Random Integers	96
Problem 346: Strong Repunits	97
Problem 347: Largest Integer Divisible by Two Primes	98
Problem 348: Sum of a Square and a Cube	99
Problem 357: Prime Generating Integers	100
Problem 587: Concave Triangle	101
Problem 607: Marsh Crossing	102
Problem 686: Powers of Two with Leading Digits	103
Problem 719: Number Splitting	104
Problem 788: Dominating Numbers	105
Problem 808: Reversible Prime Squares	106
Problem 836: A Bold Proposition	107
Problem 932: 2025	108

Problem 1: Multiples of 3 or 5

Solution: 233168

Add the integers below 1000 that are divisible by 3 or by 5, looping once and testing each.

Problem 2: Even Fibonacci Numbers

Solution: 4613732

Generate Fibonacci numbers up to four million and sum the even ones.

Problem 3: Largest Prime Factor

Solution: 6857

Factorise 600851475143 by trial division and take the largest prime factor.

Problem 4: Largest Palindrome Product

Solution: 906609

Check every product of two three-digit numbers and keep the largest that reads the same in reverse.

Problem 5: Smallest Multiple

Solution: 232792560

Scan upward for the first number divisible by every integer from 2 to 20.

Problem 6: Sum Square Difference

Solution: 25164150

Subtract the sum of the squares of 1 to 100 from the square of their sum, using 5050 for the latter.

Problem 7: 10001st Prime

Solution: 104743

Step through the odd numbers, testing primality, until the 10001st prime is reached.

Problem 8: Largest Product in a Series

Solution: 23514624000

Slide a window of thirteen digits across the given 1000-digit number and take the largest product.

Problem 9: Special Pythagorean Triplet

Solution: 31875000

Search for $a + b + c = 1000$ with $a^2 + b^2 = c^2$ and return the product abc .

Problem 10: Summation of Primes

Solution: 142913828922

Sum every prime below two million using a sieve.

Problem 11: Largest Product in a Grid

Solution: 70600674

For each cell and each of the eight directions, multiply four adjacent entries and keep the maximum.

Problem 12: Highly Divisible Triangular Number

Solution: 76576500

Generate triangular numbers until one has more than 500 divisors.

Problem 13: Large Sum

Solution: 5537376230

Add the fifty fifty-digit numbers and report the first ten digits.

Problem 14: Longest Collatz Sequence

Solution: 837799

Compute the Collatz chain length for every start below one million and report the longest.

Problem 15: Lattice Paths

Solution: 137846528820

The monotonic routes through a 20×20 grid number $\binom{40}{20}$.

Problem 16: Power Digit Sum

Solution: 1366

Sum the decimal digits of 2^{1000} .

Problem 17: Number Letter Counts

Solution: 21124

Count the letters used to spell every number from 1 to 1000, using small per-range letter tables (with the British "and").

Problem 18: Maximum Path Sum I

Solution: 1074

Work up the triangle from the bottom: each cell takes its value plus the larger of the two cells below, leaving the answer at the apex.

Problem 19: Counting Sundays

Solution: 171

Walk a running day count month by month from 1901 to 2000, counting the months that begin on a Sunday.

Problem 20: Factorial Digit Sum

Solution: 648

Sum the decimal digits of $100!$.

Problem 21: Amicable Numbers

Solution: 31626

Sum the amicable numbers below 10000, where $d(a)$ is the proper-divisor sum and $a \neq d(a) = b$ with $d(b) = a$.

Problem 22: Names Scores

Solution: 871198282

Sort the names alphabetically, then sum each name's position times its alphabetical letter value.

Problem 23: Non-Abundant Sums

Solution: 4179871

Sum the positive integers that cannot be written as the sum of two abundant numbers (numbers whose proper divisors sum to more than themselves). Every integer greater than 28123 can be so written, so that bound is the cap.

Sieve the proper-divisor sums for all $i \leq 28123$ (walk each i across its multiples, adding i to each). The abundant numbers are those with divisor-sum exceeding themselves. Mark every $a + b$ for abundant $a \leq b$, then add up the integers left unmarked.

Problem 24: Lexicographic Permutations

Solution: 2783915460

Generate the permutations of the digits 0 to 9 in lexicographic order and take the millionth.

Problem 25: 1000-digit Fibonacci Number

Solution: 4782

Find the index of the first Fibonacci number with one thousand digits.

Problem 26: Reciprocal Cycles

Solution: 983

For each denominator below 1000, run long division while tracking remainders; the recurring cycle length is found by the first repeated remainder. Report the longest.

Problem 27: Quadratic Primes

Solution: -59231

Search coefficients $a, b \in (-1000, 1000)$ for the quadratic $n^2 + an + b$ that produces the longest run of primes from $n = 0$, returning ab .

Problem 28: Number Spiral Diagonals

Solution: 669171001

Sum the diagonals of a 1001×1001 spiral using the closed form for each ring's four corners.

Problem 29: Distinct Powers

Solution: 9183

Count the distinct values a^b for $2 \leq a, b \leq 100$ with a set.

Problem 30: Digit Fifth Powers

Solution: 443839

Sum the numbers equal to the sum of the fifth powers of their digits (the search bound is $6 \cdot 9^5$).

Problem 31: Coin Sums

Solution: 73682

Count the ways to make two pounds from the coin denominations by recursing over the coin list.

Problem 32: Pandigital Products

Solution: 45228

Sum the distinct products c for which $a \times b = c$ and the three numbers together use each digit 1–9 exactly once. The digit counts can only split as $1 \times 4 = 4$ or $2 \times 3 = 4$, so the smaller factor is below 100. Iterate a and b , concatenate a , b and ab , and keep the product whenever those nine characters are exactly $\{1, \dots, 9\}$.

Problem 33: Digit Cancelling Fractions

Solution: 100

Find the non-trivial two-digit fractions whose value is unchanged by cancelling a shared digit, and reduce the product of the four.

Problem 34: Digit Factorials

Solution: 40730

Sum the numbers equal to the sum of the factorials of their digits.

Problem 35: Circular Primes

Solution: 55

Count the primes below one million for which every digit rotation is also prime.

Problem 36: Double-base Palindromes

Solution: 872187

Sum the numbers below one million that are palindromic in both base ten and base two.

Problem 37: Truncatable Primes

Solution: 748317

Find the eleven primes that remain prime as digits are stripped from the left and from the right.

Problem 38: Pandigital Multiples

Solution: 932718654

Search for the largest 1 to 9 pandigital formed by concatenating $n, 2n, \dots$ for some integer.

Problem 39: Integer Right Triangles

Solution: 840

For perimeters up to 1000, count integer right-triangle solutions and report the perimeter with the most.

Problem 41: Pandigital Prime

Solution: 7652413

Test the permutations using digits 1 to n for primality and keep the largest prime.

Problem 42: Coded Triangle Numbers

Solution: 162

A word's letter value is triangular exactly when $1 + 8v$ is a perfect square (for value v); count the words that qualify.

Problem 43: Sub-string Divisibility

Solution: 16695334890

Sum the 0-to-9 pandigital numbers whose sliding three-digit windows $d_2d_3d_4, d_3d_4d_5, \dots, d_8d_9d_{10}$ are divisible by 2, 3, 5, 7, 11, 13, 17 respectively. Instead of testing all $10!$ permutations, build from the right: start with the three-digit multiples of 17 that have distinct digits, then repeatedly prepend a digit so each new leading window is divisible by the next prime down the list. Finally prepend the one unused digit and sum the results.

Problem 44: Pentagon Numbers

Solution: 5482660

Search pairs of pentagonal numbers whose sum and difference are both pentagonal, minimising the difference, using the inverse formula to test pentagonality.

Problem 46: Goldbach's Other Conjecture

Solution: 5777

Find the smallest odd composite that cannot be written as a prime plus twice a square.

Problem 47: Distinct Primes Factors

Solution: 134043

Find the first run of four consecutive integers each having exactly four distinct prime factors.

Problem 48: Self Powers

Solution: 9110846700

Sum $1^1 + 2^2 + \dots + 1000^{1000}$ and report the last ten digits.

Problem 49: Prime Permutations

Solution: 296962999629

Find the arithmetic progression of three four-digit primes that are digit permutations of one another, other than the given 1487 case.

Problem 50: Consecutive Prime Sum

Solution: 997651

Track running sums of consecutive primes below one million and report the prime that is the longest such sum.

Problem 51: Prime Digit Replacements

Solution: 121313

Find the smallest prime belonging to an eight-member family formed by replacing one fixed set of digit positions with each value 0–9. Sieve the primes for constant-time tests and, for each prime, try the position masks. Only a mask replacing a number of positions divisible by 3 can produce eight primes: otherwise, as the replacement digit runs 0–9, the digit sum modulo 3 cycles and culls too many candidates as multiples of 3. Return the smallest member of the first eight-family found.

Problem 52: Permuted Multiples

Solution: 142857

Find the smallest x whose multiples $2x$ through $6x$ are all digit permutations of x .

Problem 53: Combinatoric Selections

Solution: 4075

Count the binomial coefficients $\binom{n}{r}$ with $n \leq 100$ that exceed one million.

Problem 55: Lychrel Numbers

Solution: 249

Count the numbers below 10000 that fail to produce a palindrome within fifty reverse-and-add steps.

Problem 56: Powerful Digit Sum

Solution: 972

Maximise the digit sum of a^b over $a, b < 100$.

Problem 57: Square Root Convergents

Solution: 153

Iterate the continued-fraction convergents of $\sqrt{2}$ and count those whose numerator has more digits than the denominator.

Problem 58: Spiral Primes

Solution: 26241

Grow the diagonals of a number spiral until the proportion of prime diagonal entries falls below ten percent.

Problem 62: Cubic Permutations

Solution: 127035954683

Group cubes by their sorted digits and find the smallest cube belonging to a group of exactly five digit-permutation cubes.

Problem 63: Powerful Digit Counts

Solution: 49

Count the n -digit positive integers that are also n th powers.

Problem 65: Convergents of e

Solution: 272

Build the hundredth convergent of the continued fraction for e and sum the digits of its numerator.

Problem 67: Maximum Path Sum II

Solution: 7273

The same bottom-up dynamic programming as Problem 18, applied to the hundred-row triangle read from file.

Problem 69: Totient Maximum

Solution: 510510

Maximise $n/\varphi(n)$ for $n \leq 10^6$; the optimum is the primorial 510510.

Problem 70: Totient Permutation

Solution: 8319823

We want the n with $1 < n < N$, where $N = 10^7$, that minimises the ratio $\frac{n}{\varphi(n)}$ subject to $\varphi(n)$ being a digit permutation of n .

Computing every totient at once

Factorising each n on its own is the slow part. Instead we compute φ for every value below N in a single sieve, using the product formula

$$\varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

Start with $\varphi[i] = i$ for all i . Sweep $p = 2, 3, 4, \dots$; whenever $\varphi[p]$ is still equal to p its entry has never been modified, so p must be prime. For each such p we fold in the factor $(1 - \frac{1}{p})$ across all of its multiples:

$$\varphi[k] \leftarrow \varphi[k] - \frac{\varphi[k]}{p}, \quad k = p, 2p, 3p, \dots$$

After the sweep $\varphi[i]$ holds $\varphi(i)$ for every $i < N$. The cost is $O(N \log \log N)$, versus factorising N numbers separately.

Searching for the minimum ratio

Two observations keep the search cheap.

First, a digit permutation leaves the digit sum unchanged, and every integer is congruent to its digit sum modulo 9. So a necessary condition for $\varphi(n)$ to be a permutation of n is

$$n \equiv \varphi(n) \pmod{9},$$

which rejects roughly $\frac{8}{9}$ of all candidates before any further work. For the survivors we confirm a genuine permutation by tallying digits: increment a length-10 array once per digit of n , decrement it once per digit of $\varphi(n)$, and accept only if every entry is back to zero.

Second, the ratios are compared exactly, without floating point. The best ratio so far is kept as a fraction $n_{\text{best}}/d_{\text{best}}$, and a new candidate $n/\varphi(n)$ improves on it precisely when

$$n \cdot d_{\text{best}} < n_{\text{best}} \cdot \varphi(n),$$

a comparison of two integer products.

The search itself assumes nothing about the form of the answer, but the winner turns out to be $8319823 = 2339 \times 3557$, a product of two primes straddling $\sqrt{N} \approx 3162$. That is exactly where $\frac{n}{\varphi(n)} = \prod_{p|n} p/(p-1)$ is smallest: few prime factors, each as large as possible, while still leaving room for $\varphi(n)$ to be a permutation of n .

Problem 71: Ordered Fractions

Solution: 428570

Scan denominators up to one million for the reduced fraction immediately below $\frac{3}{7}$.

Problem 72: Counting Fractions

Solution: 303963552391

We count the reduced proper fractions n/d with $1 \leq n < d$, $\gcd(n, d) = 1$, and $d \leq N$ for $N = 10^6$.

Fix a denominator d . The valid numerators are exactly the integers in $[1, d)$ that are coprime to d , and the count of those is by definition Euler's totient $\varphi(d)$. Summing over every denominator gives

$$\sum_{d=2}^N \varphi(d).$$

(The denominator $d = 1$ contributes nothing, since $n < d$ leaves no numerator.)

So the whole problem reduces to one totient sieve: build $\varphi(d)$ for all $d \leq N$ with the same $O(N \log \log N)$ sieve used in Problem 70, then add up the entries from 2 to N .

Problem 73: Counting Fractions in a Range

Solution: 7295372

Count the reduced fractions strictly between $\frac{1}{3}$ and $\frac{1}{2}$ with denominator at most 12000.

Problem 74: Digit Factorial Chains

Solution: 402

Count the starting numbers below one million whose digit-factorial chain has exactly sixty non-repeating terms.

Problem 76: Counting Summations

Solution: 190569291

Count the ways to write 100 as a sum of two or more positive integers, via memoised recursion over the largest part.

Problem 77: Prime Summations

Solution: 71

Find the first value expressible as a sum of primes in more than five thousand ways.

Problem 80: Square Root Digital Expansion

Solution: 40886

For each non-square up to 100, compute one hundred digits of its square root with integer arithmetic and sum them.

Problem 85: Counting Rectangles

Solution: 2772

Search grid dimensions for the rectangle count nearest two million; a $w \times h$ grid contains $\sum (w - i)(h - j)$ sub-rectangles.

Problem 87: Prime Power Triples

Solution: 1097343

Count the numbers below 5×10^7 expressible as $p^2 + q^3 + r^4$ for primes p, q, r . The roots are small ($r \leq 84, q \leq 368, p \leq 7071$), so sieve the primes once and mark every reachable sum in a boolean array, breaking each loop as soon as the partial sum passes the limit. The answer is the number of distinct marked sums.

Problem 92: Square Digit Chains

Solution: 8581146

Every chain under repeated squared-digit-sum ends at 1 or 89; count the starts below ten million that reach 89.

Problem 95: Amicable Chains

Solution: 14316

Find the smallest member of the longest amicable chain whose every element stays at or below 10^6 (a chain is $n \rightarrow s(n) \rightarrow s(s(n)) \rightarrow \dots \rightarrow n$, where s sums proper divisors).

Sieve $s(i)$ for all $i \leq 10^6$, then from each start follow the sequence. A walk that drops below the start (its true minimum was already handled) or climbs above 10^6 is discarded; one that returns to the start is a chain. A generous length cap stops the walk from spinning forever inside a foreign cycle whose minimum exceeds the start. Track the longest.

Problem 97: Large Non-Mersenne Prime

Solution: 8739992577

Report the last ten digits of $28433 \cdot 2^{7830457} + 1$.

Problem 99: Largest Exponential

Solution: 709

Compare the base-exponent pairs from file by exponent times $\ln$ of base and report the line of the largest.

Problem 104: Pandigital Fibonacci Ends

Solution: 329468

Find the Fibonacci index whose first nine and last nine digits are each 1 to 9 pandigital, keeping the tail modulo 10^9 to stay fast.

Problem 108: Diophantine Reciprocals I

Solution: 180180

We want the least n for which $1/x + 1/y = 1/n$ (positive integers) has more than 1000 distinct solutions.

Counting solutions

Clearing denominators turns $1/x + 1/y = 1/n$ into $xy = nx + ny$, and adding n^2 to both sides factors it (Simon's favourite factoring trick):

$$(x - n)(y - n) = n^2.$$

Every factorisation $n^2 = uv$ with $u \leq v$ gives a solution $x = n + u$, $y = n + v$, so the number of unordered solutions is

$$\frac{d(n^2) + 1}{2},$$

where d counts divisors. The $+1$ then halving accounts for the single symmetric pair $u = v = n$. Requiring more than 1000 solutions means

$$d(n^2) \geq 2001.$$

Finding the smallest such n

Write $n = \prod_i p_i^{a_i}$. Then $n^2 = \prod_i p_i^{2a_i}$ and

$$d(n^2) = \prod_i (2a_i + 1).$$

So we need the smallest integer whose exponents satisfy $\prod (2a_i + 1) \geq 2001$. For a fixed multiset of exponents the product $\prod p_i^{a_i}$ is smallest when the largest exponents sit on the smallest primes, so it suffices to search exponent tuples that are non-increasing across $2, 3, 5, 7, \dots$

A short depth-first search does exactly this: at prime p_i try each exponent a (capped by the previous prime's exponent), divide the remaining divisor target by $(2a + 1)$, and recurse. The search space is tiny and returns instantly. The minimum is

$$n = 180180 = 2^2 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13, \quad d(n^2) = 5 \cdot 5 \cdot 3 \cdot 3 \cdot 3 \cdot 3 = 2025,$$

giving $(2025 + 1)/2 = 1013$ solutions.

Problem 112: Bouncy Numbers

Solution: 1587000

Count bouncy numbers (digits neither wholly non-decreasing nor non-increasing) until they make up ninety-nine percent of all numbers seen.

Problem 114: Counting Block Combinations I

Solution: 16475640049

A special case of Problem 115 with minimum block length three.

Problem 115: Counting Block Combinations II

Solution: 168

Count the arrangements of red blocks of length at least m separated by single gaps via memoised recursion, then find the row length where the count first passes one million.

Problem 116: Red, Green or Blue Tiles

Solution: 20492570929

Count the ways to lay tiles of a single length (2, 3 or 4) on a row of fifty squares, summed over the three colours, excluding the empty arrangement.

Problem 118: Pandigital Prime Sets

Solution: 44680

We count the distinct sets of primes whose digits, taken together, are exactly 1, 2, ..., 9 each used once (for example {2, 5, 47, 89, 631}).

Two stages: count, then combine

Such a set is a partition of the nine digits into blocks, where each block's digits are arranged into a prime. Crucially, a set of primes determines its own partition (group the primes by which digits they use), so we may count

(set partitions of $\{1, \dots, 9\}$) $\times$ (a prime arrangement chosen per block)

without any risk of double counting, as long as each partition is enumerated once.

Stage 1 — primes per digit subset

For every non-empty subset S of the digits, let $f(S)$ be the number of permutations of S that are prime. Two cheap rules cut the work sharply:

- if $|S| > 1$ and the digit sum of S is divisible by 3, then every arrangement is divisible by 3, so $f(S) = 0$;
- a prime longer than one digit must end in 1, 3, 7 or 9, so only those arrangements are tested.

We store $f(S)$ for all 511 subsets, indexing each subset by a 9-bit mask.

Stage 2 — combining blocks

Let $g(M)$ be the number of distinct prime sets using exactly the digits in mask M .

Splitting off the block that contains the lowest set bit makes every partition unique:

$$g(M) = \sum_{B \subseteq M, \ell \in B} f(B) \cdot g(M \setminus B), \quad g(0) = 1,$$

where ℓ is the lowest set bit of M and B ranges over sub-masks containing it.

Enumerating sub-masks costs $O(3^9)$ overall. The answer is g of the full mask, $g(111111111_2) = 44680$.

Problem 123: Prime Square Remainders

Solution: 21035

Using $(p-1)^n + (p+1)^n \bmod p^2$, find the first prime index whose remainder exceeds 10^{10} .

Problem 124: Ordered Radicals

Solution: 21417

Sort 1 to 100000 by radical (the product of distinct prime factors), breaking ties by value, and read off the 10000th.

Problem 125: Palindromic Sums

Solution: 2906969179

Sum the palindromes below 10^8 that can be written as a sum of two or more consecutive squares. Rather than test every number, generate the sums: for each starting base accumulate consecutive squares until the running total reaches 10^8 , recording each palindromic total in a set so that a number expressible in several ways is still counted once. Sum the set.

Problem 145: Reversible Numbers

Solution: 608720

A number n is *reversible* if $n + \text{reverse}(n)$ has only odd decimal digits, with no leading zero in either n or $\text{reverse}(n)$. We count reversible $n < 10^9$.

Instead of testing a billion numbers, count by digit length d . Write the digits of n as $a_1 a_2 \dots a_d$. When n and $\text{reverse}(n)$ are added, column j (counting from the right) always pairs the same two digits, $a_j + a_{d+1-j}$, so the column sums are symmetric. Let $S_j = a_j + a_{d+1-j}$, and let carries propagate right to left.

Even length d

The units column gives $a_1 + a_d = S_1$; for that output digit to be odd, S_1 must be odd. The top column outputs $S_1 + c$ where c is its incoming carry; since S_1 is already odd, c must be even, i.e. 0. Pushing this inward forces **every** column sum to be odd and below 10, so there are no carries anywhere. Counting digit pairs (x, y) with $x + y$ odd and $x + y < 10$:

- the outer pair needs $x, y \geq 1$ (no leading zeros): 20 choices;
- each of the remaining $d/2 - 1$ pairs allows a zero: 30 choices each.

So an even length contributes $20 \cdot 30^{d/2-1}$.

Odd length d

Now there is a middle column that adds a digit to itself, $2a_m$, which is even. Its output can only be odd if a carry of 1 arrives from the right. That single required carry cannot coexist with the carry-free pattern above unless the lengths line up: the consistency of the inward and outward carries closes only when $d \equiv 3 \pmod{4}$. Those lengths contribute $100 \cdot 500^{(d-3)/4}$ (for $d = 3$ this is the $a_1 + a_d \in \{11, 13, 15, 17\}$ outer pair, 20 ways, times the 5 middle digits 0..4, giving 100); lengths with $d \equiv 1 \pmod{4}$ contribute nothing.

Total

Summing over $d = 1, \dots, 9$ (the digit lengths below 10^9):

$$\underbrace{20 + 600 + 18000 + 540000}_{\text{even}} + \underbrace{100 + 50000}_{d \equiv 3} = 608720.$$

A direct brute-force count confirms each per-length value (for instance 120 reversible numbers below 1000).

Problem 173: Square Laminae

Solution: 1572729

Count the square laminae (hollow square frames) that can be built with at most 10^6 tiles. A frame of outer width a and hole width b (same parity, $b < a$) uses $a^2 - b^2$ tiles. For each a , start from the widest hole $b = a - 2$ and shrink it; the tile count rises monotonically, so stop the moment it exceeds 10^6 . The outer loop ends once even the thinnest frame, costing $4a - 4$ tiles, no longer fits.

Problem 179: Consecutive Positive Divisors

Solution: 986262

We count integers $1 < n < 10^7$ for which n and $n + 1$ have the same number of divisors.

Computing $d(n)$ by factorising each n is wasteful. Instead sieve all divisor counts at once: for every i from 1 to N , walk its multiples $i, 2i, 3i, \dots$ and add one to each, since i divides every one of them. After the sweep $d[k]$ holds the divisor count of k . The work is the harmonic sum

$$\sum_{i=1}^N \frac{N}{i} \approx N \ln N,$$

versus a per-number factorisation. A compact `int32` array keeps the strided increments cache-friendly.

Then a single linear scan counts the positions where $d[n] = d[n + 1]$ for $2 \leq n \leq N - 1$, giving 986262.

Problem 187: Semiprimes

Solution: 17427258

We count the composite numbers $n < N$ (with $N = 10^8$) that are a product of exactly two primes, $n = pq$ with $p \leq q$.

Writing each such n with its smaller factor first makes the count a sum over p : for a fixed prime p , the partners are the primes q with $p \leq q$ and $pq < N$, i.e. $p \leq q \leq \lfloor (N - 1)/p \rfloor$. Counting unordered pairs this way (insisting $q \geq p$) visits each semiprime once. Only primes with $p^2 < N$, that is $p < \sqrt{N} \approx 10^4$, can be the smaller factor, so there are barely more than a thousand outer terms.

The largest partner occurs at $p = 2$, where q can reach $(N - 1)/2$, so we sieve every prime up to $N/2$ once. For each small prime p a binary search in that sorted list returns the number of primes up to $\lfloor (N - 1)/p \rfloor$; subtracting the primes below p leaves the count of valid q . Summing gives 17427258 (and 10 below 30, matching the example).

Problem 188: Hyperexponentiation

Solution: 95962097

Compute the tetration $1777 \uparrow\uparrow 1855$ modulo 10^8 by repeated modular exponentiation from the top down.

Problem 203: Squarefree Binomial Coefficients

Solution: 34029210557338

Sum the distinct squarefree entries in the first 51 rows of Pascal's triangle. The key fact is that every prime factor of $\binom{n}{k}$ with $n \leq 50$ is at most n , so squarefreeness only needs testing against the primes up to 47 — the large binomials themselves never have to be factored. Collect the distinct values from the left half of each row (using $\binom{n}{k} = \binom{n}{n-k}$) and add those divisible by no p^2 .

Problem 301: Nim

Solution: 2178309

Count the $n \leq 2^{30}$ with $n \oplus 2n \oplus 3n = 0$, the losing positions of this three-heap Nim.

Problem 315: Digital Root Clocks

Solution: 13625242

Two seven-segment clocks each display a number, then repeatedly its digit sum down to a single digit, beginning and ending blank. Sam's clock clears and relights every segment at each step; Max's only toggles the segments that change. The answer is Sam's total segment transitions minus Max's, summed over the primes in $[10^7, 2 \cdot 10^7)$.

Sieve that prime range rather than testing each number for primality, and compute both transition counts per prime with integer digit arithmetic. For a step $a \rightarrow b$, Sam's cost is the total lit-segment count of a plus that of b , while Max's cost is the sum over aligned digit positions of the population count of the segment-pattern XOR (a blank screen contributing the all-off pattern).

Problem 323: Bitwise-OR Operations on Random Integers

Solution: 6.3551758451

Each step ORs in a random 32-bit integer; the expected number of steps is $\sum i \cdot p(i)$, where $p(i)$ comes from differencing the closed-form chance that all bits are set by step i .

Problem 346: Strong Repunits

Solution: 336108797689259276

Collect every number that is a repunit of length at least three in some base up to $\sqrt{n}$, together with all values that are length-two repunits, and sum the distinct ones below 10^{12} .

Problem 347: Largest Integer Divisible by Two Primes

Solution: 11109800204052

For distinct primes p, q let $M(p, q, N)$ be the largest integer $\leq N$ whose only prime factors are p and q (so $M = p^a q^b$ with $a, b \geq 1$), or 0 if none exists. $S(N)$ sums all the distinct values; we need $S(10^7)$.

Each value $p^a q^b$ has a unique prime pair, so summing distinct values is the same as summing $M(p, q, N)$ over the pairs themselves. Iterating n and factorising it is wasteful; instead iterate the pairs. A valid pair needs $pq \leq N$, and since $p < q$ this forces $p < \sqrt{N}$, so only a few hundred primes can be the smaller factor. Sieve the primes up to $N/2$ (the largest a factor can be, paired with 2), then loop p over primes below $\sqrt{N}$ and q over larger primes while $pq \leq N$.

For a fixed pair, M is found by trying each power p^a with $p^a q \leq N$ and, for each, taking the largest power $q^b \leq N/p^a$; the maximum product over all a is $M(p, q, N)$. Summing gives $S(100) = 2262$ as a check and $S(10^7) = 11109800204052$.

Problem 348: Sum of a Square and a Cube

Solution: 1004195061

We want the five smallest palindromes expressible as $a^2 + b^3$, with $a, b > 1$, in **exactly** four ways, and their sum.

The naive route, testing every integer for palindromicity, wastes almost all of its time: only about 10^5 palindromes lie below 10^9 . So we generate palindromes directly and in increasing order. A palindrome is fixed by its first half: mirror a root r of h digits to get an even-length palindrome $r \cdot 10^h + \text{reverse}(r)$, or mirror r minus its last digit for odd length. Iterating length upward, and root upward within each length, yields every palindrome in ascending order, so the first five that qualify are the smallest five.

For each palindrome p we count the representations by trying every cube b^3 with $b \geq 2$ and $b^3 + 4 \leq p$ (the $+4$ leaves room for $a^2 \geq 2^2$), and testing whether the remainder $p - b^3$ is a perfect square. Because the remainder is at least 4, its root is automatically ≥ 2 , so both parts exceed 1 as required. Each palindrome costs $O(p^{1/3})$ checks, and we keep only those with a count of exactly four.

The qualifying palindromes are 5229225 (the example), 37088073, 56200265, 108909801 and 796767697, summing to 1004195061.

Problem 357: Prime Generating Integers

Solution: 1739023853137

Sum the integers $n \leq 10^8$ for which $d + n/d$ is prime for every divisor d of n . Taking $d = 1$ forces $n + 1$ to be prime, so n is even and the only candidates are $n = p - 1$ for primes p . Sieve the primes up to 10^8 for constant-time primality tests, then for each candidate check the divisor pairs with $d \leq \sqrt{n}$ (the condition is symmetric in d and n/d), bailing out at the first failure — usually already at $d = 2$.

Problem 587: Concave Triangle

Solution: 2240

Let the bottom-left of the rectangle be at $(0, 0)$ and the circles to have radius 1.

The area of the L-section is $1 - \frac{\pi}{4}$.

The diagonal line has slope $\frac{1}{n}$, and the intersection of the line with the first circle has coordinate:

$$x = \frac{-\sqrt{2} \cdot n^{\frac{3}{2}} + n^2 + n}{n^2 + 1}$$
$$y = \frac{x}{n}$$

The concave triangle can be split into a triangle and a curved section.

The area of the triangle is $\text{TriangleArea} = \frac{1}{2}xy$.

Let:

$$\text{CircleInt}(a, b) = \int_a^b (\sqrt{1 - x^2}) \, dx$$
$$= \frac{1}{2} \left(x\sqrt{1 - x^2} + \sin^{-1}(x) \right) \Big|_a^b$$

Then the area of the curved section is $\text{SectionArea} = (1 - x) - \text{CircleInt}(0, 1 - x)$.

And so the ratio is:

$$\frac{\text{TriangleArea} + \text{SectionArea}}{1 - \frac{\pi}{4}}$$

Problem 607: Marsh Crossing

Solution: 13.1265108586

Minimise the crossing time over a diagonal marsh of banded speeds by binary-searching the entry angle and applying Snell's law at each band boundary.

Problem 686: Powers of Two with Leading Digits

Solution: 193060223

Using the fractional part of $j \cdot \log_{10} 2$ to detect a leading block of 123, count the powers of two that start with 123 until the 678910th.

Problem 719: Number Splitting

Solution: 128088830547982

An *S-number* is a perfect square n whose decimal digits can be cut into two or more contiguous blocks that add up to $\sqrt{n}$ (for example 6724, since $6 + 72 + 4 = 82 = \sqrt{6724}$).

We want $T(N) = \sum_{n \leq N, n \text{ is an S-number}} n$ for $N = 10^{12}$.

Rather than scan every n , iterate over the root r from 2 to $\sqrt{N} = 10^6$ and test whether r^2 splits to r . The test is a pruned recursion. Splitting a number can only lower its value-sum (the maximum, one block, is the number itself), so for a target t and remaining number m :

- if $m < t$ it is impossible;
- if $m = t$ the remaining number is itself the final block, success;
- otherwise peel off a trailing block $r = m \bmod 10^k$, and if $r < t$ recurse on the prefix $\lfloor m/10^k \rfloor$ with target $t - r$.

The trailing-block extraction handles blocks with leading zeros automatically (so $9801 = 98 + 0 + 1$ is found). Because $r^2 \neq r$ for $r \geq 2$, any successful split uses at least two blocks, as required. As a check, $T(10^4) = 41333$.

Problem 788: Dominating Numbers

Solution: 471745499

A number is *dominating* when one digit value occupies more than half of its places (for example 2022, where three of four digits are 2). Let $D(N)$ count dominating numbers below 10^N ; we need $D(2022) \bmod (10^9 + 7)$.

Counting by length

Fix a digit length d . If a value v fills more than half the d places then no other value can also exceed half, so each dominating number has a **unique** dominant value. We may therefore sum over the count k of that value with $k > d/2$ and never double count.

For a fixed value v appearing exactly k times the placements split by whether v is the leading digit, and there is a separate correction for the leading-zero rule when $v = 0$. Writing all of that out and summing over the ten possible values, the corrections cancel and leave a compact form:

$$A(d) = \sum_{k > d/2} \binom{d}{k} \cdot 9^{d-k+1},$$

the number of dominating numbers with exactly d digits. Then $D(N) = \sum_{d=1}^N A(d)$. As a check, $D(4) = 603$ and $D(10) = 21893256$, matching the problem.

Evaluating modulo $10^9 + 7$

The earlier approach evaluated $\binom{d}{k}$ and 9^{d-k+1} as exact integers; at $d \approx 2000$ those are numbers with hundreds to thousands of digits, which is slow. Working modulo the prime $p = 10^9 + 7$ keeps every value small: precompute factorials and their modular inverses (via Fermat's little theorem) up to N , and the powers of 9 up to $N + 1$. The double sum is then $O(N^2)$ cheap modular multiplications, giving $D(2022) \equiv 471745499 \pmod{p}$.

Problem 808: Reversible Prime Squares

Solution: 3807504276997394

A reversible prime square is the square of a prime that is not a palindrome and whose digit reversal is also the square of a prime. Because squares grow with their root, iterating primes in increasing order yields these in increasing order too. Sieve primes up to about $3.2 \cdot 10^7$ (where the fiftieth occurs); for each prime p , reverse p^2 , and if the reversal is a perfect square whose root is prime, count p^2 . Stop at the fiftieth and sum.

Problem 836: A Bold Proposition

Solution: aprilfoolsjoke

An April Fools' problem; the answer is the literal string hidden in the prompt.

Problem 932: 2025

Solution: 72673459417881349

A 2025-number N splits into a leading part a and a trailing part b (with b having a fixed digit length and no leading zero) such that $N = (a + b)^2$. Writing $s = a + b$, every such N is simply s^2 , so iterate over s rather than over the parts: for each split position k , set $a = \lfloor N/10^k \rfloor$ and $b = N \bmod 10^k$, and accept when $a + b = s$ and b has exactly k digits.

A cheap filter cuts the search by more than four-fifths. Since $10^k \equiv 1 \pmod{9}$, we have $N = a \cdot 10^k + b \equiv a + b = s \pmod{9}$; combined with $N = s^2$ this forces $s^2 \equiv s$, i.e. $s \equiv 0$ or $1 \pmod{9}$. Only those s need be examined.